

OVERVIEW

The DCC1000 is a single-channel, isolated VINT motor controller for one brushed DC motor or DC linear actuator. It accepts an 8-30 V DC motor supply and can deliver up to 25 A continuously. It can run as a direct DC motor controller or as a closed-loop position controller when an encoder is installed.

Official source: <https://www.phidgets.com/?prodid=965>

This reference condenses the product description, user-guide notes, specifications, software channels, and device history. Always check the official page and current Phidget22 documentation before designing safety-critical or high-current equipment.

At a glance

Item	Value
Product	DC Motor Phidget
SKU	DCC1000_0
Motor type	Brushed DC motor
Motor outputs	1
Motor supply	8-30 V DC
Continuous motor current	25 A maximum
Control connection	VINT (VINT Hub required)
Driver support	Phidget22
Operating modes	DCMotor; MotorPositionController
PWM frequency	25 kHz (ultrasonic)
Encoder inputs	1, quadrature x4
Analog feedback input	1 voltage-ratio input, 0-5 V
On-board monitoring	Motor current and board temperature

What it does

In **DCMotor** mode, software directly commands motor velocity (signed duty cycle), acceleration, and current limit. This mode is appropriate for basic speed and direction control and for applications that implement their own feedback loop.

In **MotorPositionController** mode, the built-in encoder input is used for fast closed-loop position control. Phidgets supplies tuning support through the Phidget Control Panel on Windows. An encoder is required for this mode.

The board can also read motor current, an external feedback potentiometer through its voltage-ratio input, and its own temperature. These channels are useful for diagnostics, load monitoring, linear-actuator feedback, and thermal protection.

Key features

- Up to 25 A continuous current for one brushed DC motor.
- Closed-loop position control through the built-in encoder interface.
- Electrical isolation between the controller and VINT side (5 kV DC specified isolation voltage).
- 25 kHz PWM, above the normal audible range.
- Protection features: overcurrent protection, reverse-polarity protection, thermal monitoring, and a standard fail-safe mode.

Required and recommended hardware

VINT Hub Required. The DCC1000 does not connect directly to USB; connect it to a VINT port with a three-wire Phidget cable. Compatible hosts include USB VINT Hubs, the HUB5000 Wi-Fi VINT Hub, and supported PhidgetSBC models.

Motor supply	Required. Use a regulated 8-30 V DC supply selected for the motor voltage and expected load. Phidgets recommends 12 V for smaller motors and 24 V for larger motors; the official compatibility section suggests a 5 A supply for good performance in typical applications. A design that expects higher motor current must use a suitably rated supply, wiring, connectors, and protection.
Power Guard	Required for most applications according to Phidgets. Review the current Power Guard guide for placement and sizing.
Encoder	Required only for closed-loop position control. Connect a compatible quadrature encoder to the encoder input.
Feedback potentiometer	Optional. Linear actuators with a 0-5 V feedback potentiometer can be monitored through the voltage-ratio input.

Connection and commissioning checklist

1. Keep power off while wiring. Confirm the motor is brushed DC and its rated voltage is within the controller's 8-30 V range.
2. Size conductors for the actual current, length, ambient temperature, and applicable electrical rules. The board accepts 10-26 AWG wire; this range is a terminal capability, not a blanket current-rating guarantee.
3. Connect the motor supply and motor leads to their labeled terminals. Add the recommended Power Guard and external protection for the installation.
4. Connect the DCC1000 to a VINT Hub with a three-wire Phidget cable, then connect the hub to the host computer or network.
5. Install the current Phidget22 library and open the Phidget Control Panel. Verify every channel appears before enabling the motor.
6. Set the current limit to the motor manufacturer's specified coil/rated current unless the application requires a lower protective limit.
7. Begin with a low velocity and conservative acceleration. Confirm direction, current, temperature, encoder count direction, and mechanical limits.
8. Configure a failsafe timeout and ensure the application regularly services the device. Test the safe-state behavior before normal operation.

Important operating notes

Current limit	The current limit controls how much current can be supplied to the motor. Phidgets generally advises using the motor's specified coil current. The setting resolution is 17.9 mA.
Fan mode	The on-board fan can be controlled by software. Auto mode turns it on in response to rising board temperature.
Regulator gain	Current-loop behavior depends on supply voltage and motor inductance. Large motors commonly have lower inductance, and higher supply voltage changes current more quickly. Gain set too low can allow current spikes, torque variation, and possible controller damage. Tune cautiously using Phidgets guidance.
Direction	If a position-control system moves away from its target, reverse the motor leads. A brushed DC motor can operate with reversed polarity; this changes its rotation direction. Confirm that encoder sign and motor direction form negative feedback before applying significant power.
Linear actuators	Software must prevent travel beyond the mechanical end stops. Use the actuator's feedback potentiometer, limit switches where appropriate, current monitoring, and a tested shutdown strategy.

Specifications

Board and controller

Property	Specification
Controlled by	VINT
Driver	Phidget22
Maximum VINT communication speed	10 kbit/s
Motor type / ports	DC motor / 1 port
Velocity resolution	0.001 duty cycle
Acceleration resolution	1 duty cycle/s
Acceleration range	0.5 to 10,000 duty cycle/s
Acceleration-time range	20 ms to 20 s
PWM frequency	25 kHz
Controller sampling interval	50 ms to 60 s per sample
Current-limit resolution	17.9 mA

Electrical

Property	Specification
Continuous motor current maximum	25 A
Supply voltage	8-30 V DC
VINT current, unconfigured	500 microampere
VINT current, maximum	2 mA
Power, unconfigured	288 mW
Operating power consumption	Motor power plus 700 mW
Replacement fuse	20 A slow-blow blade, regular or micro
Isolation voltage	5 kV DC

Encoder interface

Property	Specification
Encoder inputs	1
Interface resolution	x4 quadrature
Maximum count rate	400,000 pulses/s
Time resolution	1 microsecond
Sampling interval	50 ms to 60 s per sample
Input low, maximum	800 mV DC
Input high, minimum	2 V DC

Voltage-ratio input and temperature

Property	Specification
Voltage inputs	1
Input range	0-5 V DC
VoltageRatio resolution	0.00026
Maximum measurement error	0.5%
Input impedance	324 kilohm
Voltage sampling interval	500 ms to 60 s per sample
Temperature resolution	0.04 degrees C

Physical and environmental

Property	Specification
Recommended wire size	10-26 AWG
Operating temperature	-40 to 85 degrees C
Country of origin	China

Phidget22 software channels

Function	API class	Channel
Motor velocity/current/fan	DCMotor	0
Closed-loop position	MotorPositionController	0
Quadrature encoder	Encoder	0
Analog feedback	VoltageRatioInput	0
Motor-current measurement	CurrentInput	0
Board-temperature measurement	TemperatureSensor	0

The generated code samples and detailed API calls vary by language and operating system. Use the code-sample generator on the official product page for current examples. Phidgets lists broad language support, including C, C#, Python, Java, JavaScript, Swift, Visual Basic, Objective-C, LabVIEW, and Max; availability varies by channel and platform.

Safety and design cautions

- 25 A at up to 30 V can overheat undersized wiring, connectors, fuses, PCB traces, and power supplies. Design the entire current path for the expected continuous and fault current.
- Motor startup, stall, reversal, and rapid braking can produce high current and regenerative voltage. Use the Phidgets Power Guard guidance and confirm the supply can tolerate the application.
- Do not rely on software alone for personnel safety or prevention of hazardous motion. Use appropriate guards, emergency stops, limit devices, and independent power isolation.
- The specified 20 A on-board replacement fuse and the 25 A continuous motor rating are different parameters. Replace the fuse only with the specified type and investigate the fault before re-energizing.
- Maintain airflow and monitor temperature, especially near maximum current or ambient temperature. The specified operating range does not guarantee full-current operation without adequate cooling.

Device history published by Phidgets

Date	Version	Change
Aug 2017	115	Product release
Oct 2017	204	Added position control; standardized motor direction
Jan 2018	205	Fixed encoder-input issue
Mar 2018	206	Fixed duty cycle not reaching 1.0
Apr 2019	207	Fixed duty-cycle averaging during current limiting
May 2019	210	Added failsafe timer
Feb 2020	211	Fixed saturation warnings at 25 A
Apr 2022	220	Double-precision position math; failsafe reset behavior updated

Further reading

- Product description, user guide, specifications, compatibility, downloads, and current code-sample generator: <https://www.phidgets.com/?prodid=965>
- From the product page, follow the links to the DCMotor API Guide, MotorPositionController API Guide, Power Guard Guide, and DC Motor and Controller Guide.

Document status: Prepared 11 September 2026 from the official product page. Product availability, pricing, software downloads, documentation, and specifications can change; the official page is authoritative.